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Dr. Marcus Webb

Head of AI Research

Meridian Technologies Inc.

800 Innovation Drive, San Francisco, CA 94107

Dear Dr. Webb,

I am writing to express my enthusiastic interest in the Senior Machine Learning Engineer position at Meridian Technologies, as advertised on your careers page. With six years of industry experience building large-scale NLP systems and a deep passion for applied AI research, I believe I am exceptionally well suited to contribute to your team's mission of democratising access to enterprise intelligence.

In my current role at DataPulse Inc., I led the design and deployment of a Retrieval-Augmented Generation platform that serves over 2,000 enterprise users daily. The system ingests heterogeneous document types — PDFs, spreadsheets, and scanned images — and provides sub-second semantic search backed by a ChromaDB vector store. I reduced hallucination rates by 38% by introducing citation-aware prompting and a faithfulness reranker, and I decreased average query latency from 1.8 s to 340 ms through HNSW index tuning and async embedding pipelines.

Prior to DataPulse, I spent three years at the University of Washington's NLP Lab as a Research Engineer, where I co-authored two papers on dense passage retrieval published at ACL 2022 and EMNLP 2023. My work on bi-encoder fine-tuning with hard-negative mining is now integrated into the open-source DPR-Turbo library with over 4,000 GitHub stars. I hold an M.S. in Computer Science (specialisation in Machine Learning) from the University of Washington.

I am particularly excited about Meridian's recently announced Agentic Search initiative. Having built multi-step reasoning chains using LangGraph and tool-calling LLMs in production, I have first-hand insight into the reliability and latency challenges of agentic systems — and concrete strategies to address them. I would welcome the opportunity to bring this experience to your team.

Thank you for taking the time to consider my application. I have attached my CV and three references. I would be delighted to discuss how my background aligns with Meridian's goals at your earliest convenience.

Yours sincerely,

Alexandra Chen